

CH 2 - Probability Rules

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2.1 Basic Axioms and Laws

Definition – Axioms of Probability

1. The probability of the sample space occurring is $P(S) = 1$
2. Each event of happening in the sample space is $0 \leq P(A) \leq 1$
3. If $A_1, A_2, \dots$ have no overlap, then $P(A_1 \cup A_2 \cup \dots) = \sum_i P(A_i)$

Note:

1. A “or” B means $A \cup B$
2. A “and” B means $A \cap B$
3. “Not in” A means the compliment $\bar{A}$

Definition – De Morgan’s Law

1. $\overline{A \cup B} = \bar{A} \cap \bar{B}$
2. $\overline{A \cap B} = \bar{A} \cup \bar{B}$

2.2 Probability Rules

1. If $A \subseteq B$ then $P(A) \leq P(B)$
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
3. $P(A \cup B \cup C) = P(A \cup B) + P(C) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$
4. In general, $P(A_1 \cup \dots \cup A_k) = \sum_i^k P(A_i) - \sum \left(\sum_{j \neq i} P(A_i A_j) \right) + \sum \left(\sum_{i \neq j \neq k} P(A_i A_j A_k) \right) + \dots - \dots$

Proof

1. $B = (B \cap A) \cup (B \cap \bar{A})$
 $P(B) = P(B \cap A) + P(B \cap \bar{A}) = P(A) + W, W > 0 \implies P(B) > P(A)$

Definition – Mutual Exclusive

Event A and B are **mutually exclusive** if $A \cap B = \emptyset$. Non-overlapping, disjoint means the same.

If $A_1, \dots, A_k$ are mutually exclusive, $P(A_1 \cup \dots \cup A_k) = \sum_i^k P(A_i)$

Examples:

Roll two fair 12-sided dice. What is the probability at least one of them is larger than 7?

$$1 - \frac{7}{12} \cdot \frac{7}{12} = \frac{95}{144} \text{ or } \frac{5}{12} \cdot \frac{7}{12} \cdot 2 + \left(\frac{5}{12}\right)^2 = \frac{95}{144}$$

2.3 Independent Events

Definition – Independent Events

Two events A and B are independent if and only if

$$P(AB) = P(A)P(B)$$

If we can directly compute $P(A)$, $P(B)$, $P(AB)$, we can determine whether or not the events are independent.

For multiple events, $A_1, A_2, \dots, A_k$, we say they are independent if and only if

$$\begin{aligned} P(A_i A_j) &= P(A_i)P(A_j) \\ P(A_i A_j A_k) &= P(A_i)P(A_j)P(A_k) \\ &\dots \end{aligned}$$

Example:

1. Roll 2 fair 6-sided dice. Let A = first die shows 3, B = sum of dice is 7

$$P(A) = \frac{1}{6}, P(B) = \frac{1}{6}$$

$$P(AB) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

2. What about sum is 8

Let C = total on 2 dice is 8

$$P(C) = \frac{5}{36}$$

$$P(A \cap C) = \frac{1}{36} \neq \frac{1}{6} \cdot \frac{5}{36}. \text{ Thus } C \text{ and } A \text{ are not independent}$$

Note: If A and B are linearly independent, so $\bar{A} \wedge B$, $A \wedge \bar{B}$, $\bar{A} \wedge \bar{B}$ are also linearly independent.

Summary:

	Mutually Exclusive	Independent	Both
Words	No overlap	No relation	At least one event is impossible
Math	$A \cap B = \emptyset$	$P(AB) = P(A)P(B)$	$P(A) = 0 \vee P(B) = 0$